1) $\sigma = 5\times10^6$ s/m | $r = 5$ m , $t = 2\times10^{-6}$ s
$\varepsilon = 5$ F/m

$$\rho_0 = \frac{1}{\frac{4}{3}\pi r^3} \ c/m^3 = \frac{3}{4\pi r^3}$$

$$\rho = \rho_0 e^{-\frac{\sigma}{\varepsilon}t}$$

$$= \rho_0 e^{-\frac{5\times10^6}{5}\times2\times10^{-6}} \ c/m^3$$

$$= \rho_0 e^{-2} \ c/m^3$$

2 marks → till this point

Total charge remains same at $t = 0$ and $t = 2\mu s$.

$$\therefore 1 = \rho_0 e^{-2} \cdot \frac{4}{3}\pi r^3$$

$$+ \sigma_{surface} 4\pi r^2$$

$$= e^{-2} + \sigma_{surface} 4\pi r^2$$

$$\Rightarrow \sigma_{surface} = \frac{(1-e^{-2})}{4\pi \times 25} \ C/m^2$$

$$= .0028 \ C/m^2$$

[Some people may keep the answer like this: give full marks]

Q2. $q/m = 1.759 \times 10^{11}$

5 $B = 1\,T$, $v = 3.518 \times 10^{5}$ m/s

$$r = \frac{mv}{qB}$$

$$= \frac{v}{\frac{q}{m} B}$$

$$= \frac{3.518 \times 10^{5}}{1.759 \times 10^{11}} \ m$$

Deduct 1.5
for calculation
mistake, otherwise full marks.

$$= 2 \times 10^{-6} \ m$$

Q3) $\phi = \frac{K}{r^4}$

$\bar{E} = -\frac{4K}{r^5}\hat{r}$ [Some people may also write $E = -\frac{4k}{r^5}$; This is okay. But if someone puts one side is scalar and one as vector, usual rule of -1 applies]

$$\text{Energy density} = \frac{1}{2}\epsilon_0 E^2$$

$$= \frac{1}{2}\epsilon_0 \left(\frac{-4K}{r^5}\right)^2$$

$$= \frac{1}{2}\epsilon_0 \cdot \frac{16K^2}{r^{10}} = \frac{8\epsilon_0 K^2}{r^{10}}$$

[No Part marking]

Q.4) $\vec{\nabla} \times (\vec{\nabla}\phi) = 0$

Sol^n LHS $= \vec{\nabla} \times (\vec{\nabla}\phi)$

$= \vec{\nabla} \times (\vec{\nabla}\phi(x,y,z))$

$= \vec{\nabla} \times \left(\frac{\partial\phi}{\partial x}\hat{x} + \frac{\partial\phi}{\partial y}\hat{y} + \frac{\partial\phi}{\partial z}\hat{z}\right)$

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix}$$

$$= \left(\frac{\partial^2\phi}{\partial y\partial z} - \frac{\partial^2\phi}{\partial z\partial y}\right)\hat{x} - \left(\frac{\partial^2\phi}{\partial x\partial z} - \frac{\partial^2\phi}{\partial z\partial x}\right)\hat{y} + \left(\frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\phi}{\partial y\partial x}\right)\hat{z}$$

$= 0$

Q.5) (a) $\int_{-1}^{+1} (r^2+2)\nabla\cdot\left(\frac{\hat{r}}{r^2}\right) dV$

Soln $\because$ $\nabla\cdot\left(\frac{\hat{r}}{r^2}\right) = 4\pi\delta^3(r)$

No part marking

also,

$$\int_V f(r)\delta^3(r-a)\,dV = f(a)$$

$$\therefore \int_{-1}^{+1} (r^2+2)\,4\pi\delta^3(r)\,dV$$

$$= 4\pi\int_{-1}^{+1}(r^2+2)\,\delta^3(r)\,dV$$

$$= 4\pi(0+2) = 8\pi$$

Alternate Soln

We know that,

$$\int_V f(\nabla\cdot A)\,d\tau = -\int_V A\cdot(\nabla f)\,d\tau + \oint_S fA\cdot da$$

Now, using above eqn

$$\int_{-1}^{+1}\frac{\hat{r}}{r^2}\cdot\left[\nabla(r^2+2)\right]dV + \oint_S (r^2+2)\frac{\hat{r}}{r^2}\cdot da$$

No part marking

$$\because \nabla(r^2+2) = 2r\hat{r}$$

$$\therefore \int\frac{2}{r}\,dV = \int\frac{2}{r}r^2\sin\theta\,dr\,d\theta\,d\phi = 8\pi\int_0^R r\,dr = 4\pi R^2$$

On boundary of sphere where $r=R$

$$da = R^2\sin\theta\,d\theta\,d\phi\,\hat{r}$$

$$\therefore \int(R^2+2)\sin\theta\,d\theta\,d\phi = 4\pi(R^2+2)$$

Substituting all

$$\int_{-1}^{+1}(r^2+2)\nabla\cdot\left(\frac{\hat{r}}{r^2}\right)dV = -4\pi R^2 + 4\pi(R^2+2) = 8\pi$$

(b) $\int_{1}^{\infty} (r^2+2)\, \nabla \cdot \left(\frac{\hat{r}}{r^2}\right) dV$

<u>Soln</u> From 1st method :-

$$\int_{1}^{\infty} (r^2+2)\, 4\pi\, \delta^3(r)\, dV$$

$$= 4\pi \int_{1}^{\infty} (r^2+2)\, \delta^3(r)\, dV$$

$$= 0$$